

MD Tasnim Mahmud Fahim

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About Me

IoT and Robotics Engineering graduate with an interdisciplinary focus spanning IoT, robotics, embedded systems, machine learning, speech processing, intelligent automation, and software/web development. Experienced in developing sensor-based systems, autonomous robotic applications, IoT monitoring dashboards, research-oriented AI solutions, and responsive web applications. Passionate about building practical and reliable intelligent systems that integrate hardware, software, machine learning, and automation.

RESEARCH PUBLICATIONS

C=Conference, J=Journal

- [C.1] Md. Kowsar Islam, Md. Tasnim Mahmud Fahim, Jakaria Saikat Dhrobo, *Saurav Chandra Das, “**An Intelligent IoT-Enabled Automated Medicine Dispensing and Health Management System,**” *IEEE International Conference Proceedings*, 2026. DOI: 10.1109/ICECTE69292.2026.11429319

EDUCATION

Bachelor Of Science in IoT and Robotics Engineering University of Frontier Technology, Bangladesh ◦ Result: GPA 3.07 out of 4.00	September, 2026
Higher Secondary Certificate; Joypurhat Government College ◦ Result: GPA 5.00 out of 5.00	2020
Secondary School Certificate Ramdeo Bazla Government High School ◦ Result: 5.00 out of 5.00	2018

TECHNICAL SKILLS

Programming Languages: Python, C, C++.
Web Programming: HTML, CSS, JavaScript, React, Node.js.
Database: MySQL, Firebase.
AI & Data Science: OpenAI API, NumPy, Pandas, Tensorflow.
Embedded System: Arduino, ESP32, NodeMCU, Raspberry Pi.
Communication Protocol: Wifi, Bluetooth, Zigbee.
Software & Tools: MS Office, VS Code, Git etc.
Game development: Unity, C#, Gameplay Programming etc.

PROJECTS

- 1. IgnisShield — Automated Fire Detection and Extinguishing System**
 - Developed an autonomous fire-extinguishing robot using ESP32, Arduino Uno, motor driver, and sensors.
 - Implemented real-time fire detection with automated movement and extinguishing responses based on sensor inputs.
 - Added Bluetooth voice and button-based controls for manual navigation and system operation.
- 2. AlertShield — IoT Safety Monitoring Dashboard**
 - Developed a Flutter-based IoT safety monitoring application with a responsive dashboard and modular navigation.

- Implemented monitoring interfaces for flood, fire, gas, vibration, and temperature conditions.
- Built interactive actuator controls and an AI recommendation interface for displaying risk levels and safety actions.

3. Reliability-Aware Speech and IoT Framework for Dyslexia-Informed Learning Support

- Developed a reliability-aware speech and IoT framework for converting spoken reading interactions into traceable literacy-support evidence.
- Implemented ASR benchmarking, word-level alignment, reliability classification, probability calibration, and selective accept/reassess/defer decisions using Python and machine learning.
- Built a traceable learner evidence system with an activity library, rule-based recommendation engine, append-only evidence storage, offline-first IoT communication, and role-aware dashboard prototype.

4. MediSage — AI-Powered Medical Information Assistant

- Developed a responsive AI-powered web application using Next.js, React, TypeScript, and Tailwind CSS.
- Integrated Gemini API for AI-powered responses and implemented user authentication using NextAuth.js.
- Used MongoDB to manage user data and chat history, with a focus on clean and reusable UI components.

5. Social Media Studio — Multi-Platform Social Campaign Publisher

- Developed a React and Express-based social media campaign publishing platform with a modular dashboard.
- Implemented automated image processing, AI-assisted caption generation, campaign scheduling, and platform-specific content adaptation.
- Built a durable SQLite-backed publishing queue with secure HMAC-SHA256 webhook verification and resilient platform delivery handling.

6. Campus SafeWitness — Anonymous Campus Incident Reporting Portal

- Developed a Next.js and TypeScript-based anonymous incident reporting portal for university campuses.
- Implemented secure incident management with Prisma ORM, PostgreSQL, JWT authentication, and role-based administration.
- Designed a responsive interface using Tailwind CSS and Shadcn/ui for streamlined incident submission and management.

EXTRA-CURRICULAR ACTIVITIES

Executive Member UFTB Robotics Club	Jan 2024 – Jan 2025
Executive Member UFTB Cybernetics Club	Jan 2023 – Sep 2024
Event Organizer / Volunteer ICT Olympiad Bangladesh Season 3	2026
Volunteer GST Examination	2023, 2024

ADDITIONAL INFORMATION

Research Interests: AI, Machine Learning, Deep Learning, IoT, Embedded System, Communication Network.

Languages: Bangla (Native), English.

REFERENCES

1. **Farzana Akter**
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2. **Md. Rafiqul Islam**
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